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[FZT658](#)

EN

This Datasheet is presented by
the manufacturer

DE

Dieses Datenblatt wird vom
Hersteller bereitgestellt

FR

Cette fiche technique est
présentée par le fabricant

400V NPN HIGH VOLTAGE TRANSISTOR IN SOT223
Features

- $BV_{CEO} > 400V$
- $I_C = 500mA$ High Continuous Current
- $I_{CM} = 1A$ Peak Pulse Current
- Low Saturation Voltage $V_{CE(SAT)} < 250mV @ 50mA$
- $h_{FE} > 40$ Specified up to 200mA for High Current Gain Hold Up
- Complementary PNP Type: FZT758
- **Lead-Free Finish; RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**

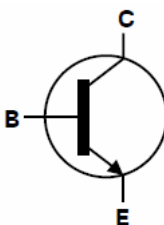
Mechanical Data

- Case: SOT223
- Case material: molded plastic. "Green" molding compound.
- UL Flammability Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Finish - Matte Tin Plated Leads, Solderable per MIL-STD-202, Method 208 Ⓔ
- Weight: 0.112 grams (approximate)

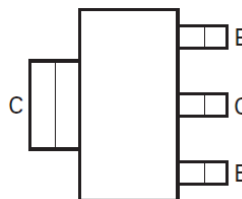
SOT223



Top View



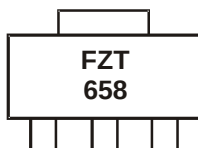
Device Symbol


 Top View
 Pin-Out

Ordering Information (Notes 4)

Product	Marking	Reel Size (inches)	Tape Width (mm)	Quantity per Reel
FZT658TA	FZT658	7	12	1,000

- Notes:
1. EU Directive 2002/95/EC (RoHS) & 2011/65/EU (RoHS 2) compliant. All applicable RoHS exemptions applied.
 2. See http://www.diodes.com/quality/lead_free.html for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
 3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
 4. For packaging details, go to our website at <http://www.diodes.com/products/packages.html>.

Marking Information


FZT658 = Product type Marking Code

Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V _{CBO}	400	V
Collector-Emitter Voltage	V _{CEO}	400	V
Emitter-Base Voltage	V _{EBO}	7	V
Continuous Collector Current	I _C	0.5	A
Peak Pulse Current	I _{CM}	1	A

Thermal Characteristics (@T_A = +25°C, unless otherwise specified.)

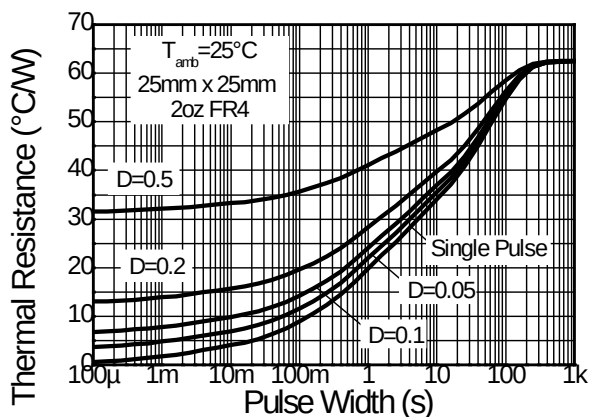
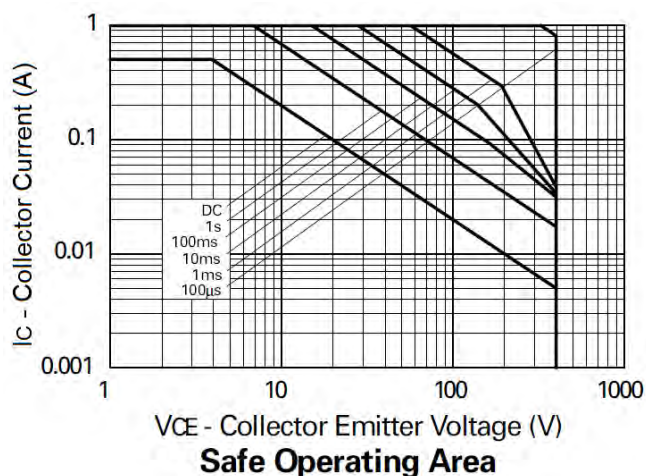
Characteristic	Symbol	Value	Unit
Power Dissipation	P _D	2	W
		3	W
Thermal Resistance, Junction to Ambient	R _{JA}	62.5	°C/W
		41.7	°C/W
Thermal Resistance, Junction to Leads (Note 7)	R _{JL}	12.93	°C/W
Operating and Storage Temperature Range	T _J , T _{STG}	-55 to +150	°C

ESD Ratings (Note 8)

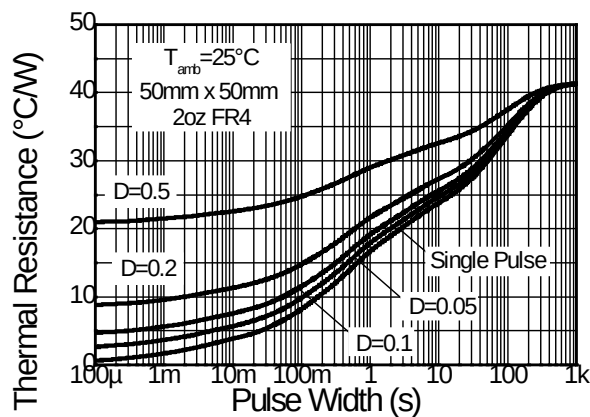
Characteristic	Symbol	Value	Unit	JEDEC Class
Electrostatic Discharge - Human Body Model	ESD HBM	• 4,000	V	3A
Electrostatic Discharge - Machine Model	ESD MM	• 400	V	C

- Notes:
- For a device surface mounted on 25mm X 25mm FR4 PCB with high coverage of single sided 1 oz copper, in still air conditions; device measured when operating in steady state condition.
 - Same as note (5), except the device is mounted on 50mm X 50mm single sided 2oz weight copper.
 - Thermal resistance from junction to solder-point (at the end of the collector lead).
 - Refer to JEDEC specification JESD22-A114 and JESD22-A115.

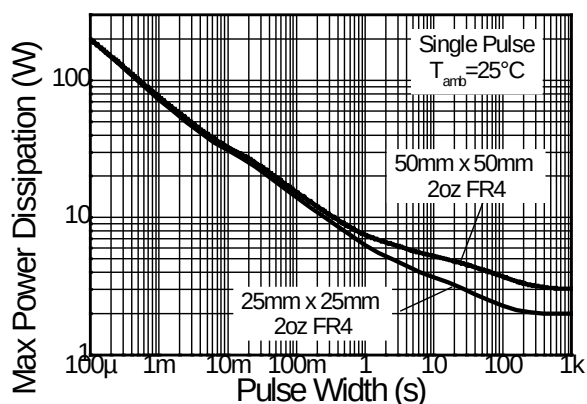
Thermal Characteristics and Derating Information



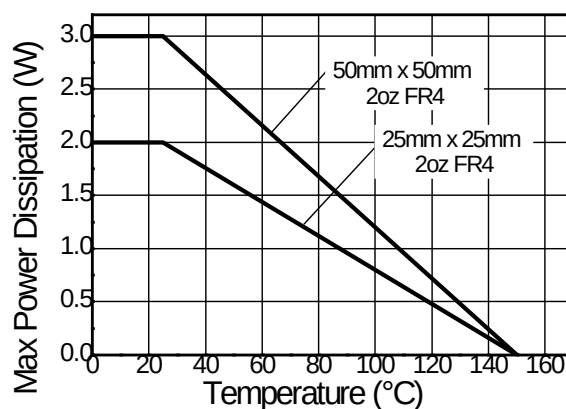
Transient Thermal Impedance



Transient Thermal Impedance



Pulse Power Dissipation



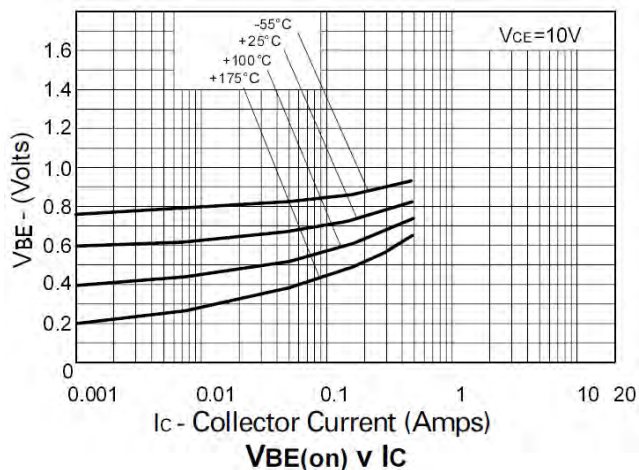
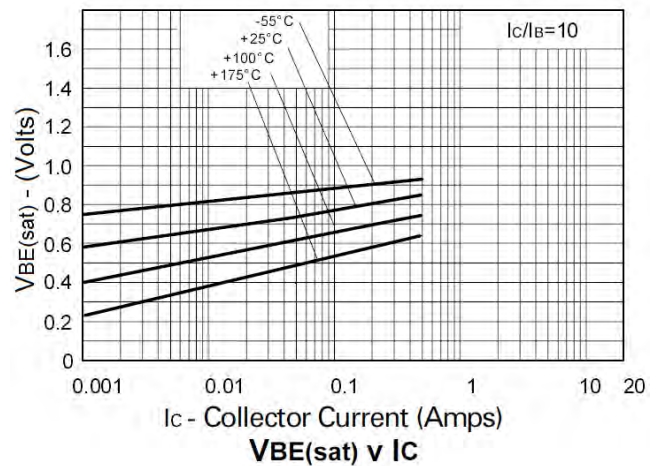
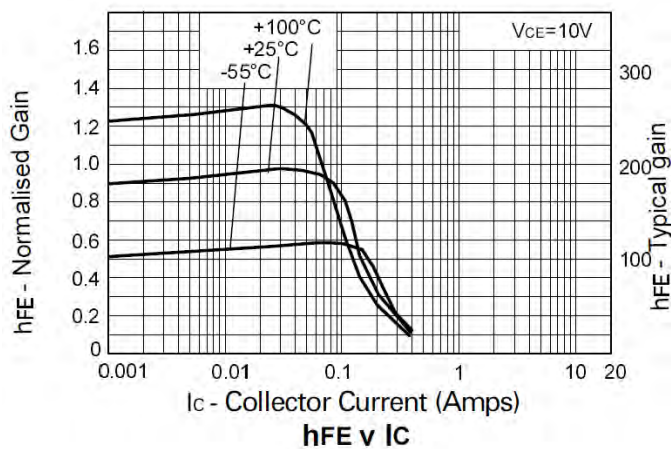
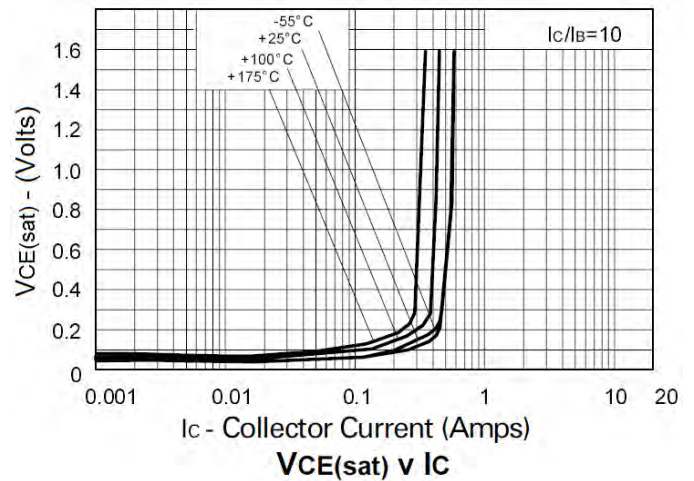
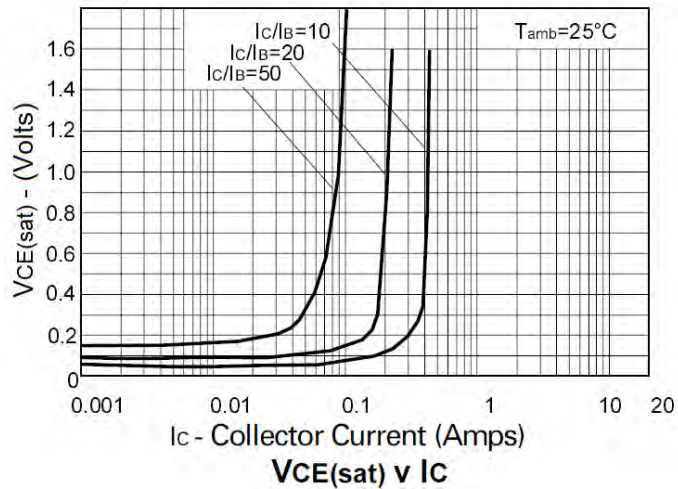
Derating Curve

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
Collector-Base Breakdown Voltage	BV _{CBO}	400	–	–	V	I _C = 100μA
Collector-Emitter Breakdown Voltage (Note 9)	BV _{CEO}	400	–	–	V	I _C = 1mA
Emitter-Base Breakdown Voltage	BV _{EBO}	7	–	–	V	I _E = 100μA
Collector Cut-off Current	I _{CBO}	–	–	100	nA	V _{CB} = 320V
Emitter Cut-off Current	I _{EBO}	–	–	100	nA	V _{EB} = 6V
Collector-Emitter Saturation Voltage (Note 9)	V _{CE(sat)}	–	–	0.30	V	I _C = 20mA, I _B = 1mA
				0.25		I _C = 50mA, I _B = 5mA
				0.50		I _C = 100mA, I _B = 10mA
Base-Emitter Saturation Voltage (Note 9)	V _{BE(sat)}	–	–	0.9	V	I _C = 100mA, I _B = 10mA
Base-Emitter Turn-On Voltage (Note 9)	V _{BE(on)}	–	–	1.0	V	I _C = 100mA, V _{CE} = 5V
DC Current Gain (Note 9)	h _{FE}	50	–	–	–	I _C = 1mA, V _{CE} = 5V
		50	–	–		I _C = 100mA, V _{CE} = 5V
		40	–	–		I _C = 200mA, V _{CE} = 10V
Current Gain-Bandwidth Product (Note 9)	f _T	50	–	–	MHz	V _{CE} = 20V, I _C = 10mA, f = 20MHz
Output Capacitance (Note 9)	C _{obo}	–	10	–	pF	V _{CB} = 20V, f = 1MHz
Switching Times	t _{on}	–	130	–	ns	I _C = 100mA, V _{CC} = 100V I _{B1} = 10mA, I _{B2} = -20mA
	t _{off}	–	3,300	–		

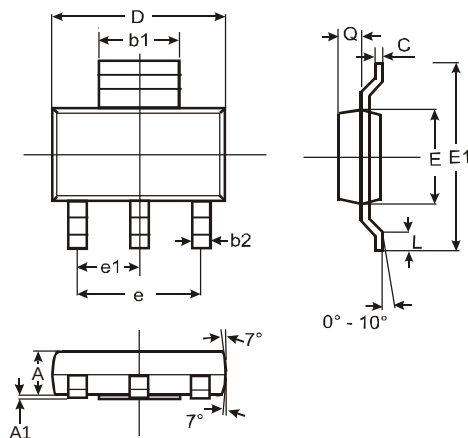
Note: 9. Measured under pulsed conditions. Pulse width • 300μs. Duty cycle • 2%

Typical Electrical Characteristics (@ $T_A = +25^\circ\text{C}$, unless otherwise specified.)



Package Outline Dimensions

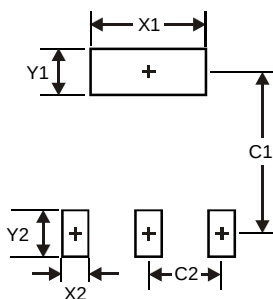
Please see AP02002 at <http://www.diodes.com/datasheets/ap02002.pdf> for latest version.



SOT223			
Dim	Min	Max	Typ
A	1.55	1.65	1.60
A1	0.010	0.15	0.05
b1	2.90	3.10	3.00
b2	0.60	0.80	0.70
C	0.20	0.30	0.25
D	6.45	6.55	6.50
E	3.45	3.55	3.50
E1	6.90	7.10	7.00
e	—	—	4.60
e1	—	—	2.30
L	0.85	1.05	0.95
Q	0.84	0.94	0.89
All Dimensions in mm			

Suggested Pad Layout

Please see AP02001 at <http://www.diodes.com/datasheets/ap02001.pdf> for the latest version.



Dimensions	Value (in mm)
X1	3.3
X2	1.2
Y1	1.6
Y2	1.6
C1	6.4
C2	2.3

Note: For high voltage applications, the appropriate industry sector guidelines should be considered with regards to voltage spacing between terminals.

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